

# Code Layout, Readability & Reusability

Project: **Auto Ticket Classification using Flow Design**

## Code Layout

| Element        | Details                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| File structure | Single module ticket_flow.py with clear sections: constants, training data, helper functions, main flow, demo run |
| Constants      | CONFIDENCE_THRESHOLD, TEAM_ROUTES and URGENT_WORDS are at the top so they can be changed without touching logic   |
| Flow order     | preprocess -> classify -> detect_priority -> process_ticket (decision) -> notify                                  |
| Entry point    | if __name__ == "__main__" block runs a demo, so the file can also be imported as a module                         |

## Readability

- Meaningful names: preprocess, classify, detect\_priority, process\_ticket, notify.
- Each function does one job; key functions have short docstrings.
- Type hints on function signatures show expected inputs and outputs.
- Consistent PEP 8 style: 4-space indentation, snake\_case names, UPPER\_CASE constants.

## Reusability

| Component        | How it can be reused                                                              |
|------------------|-----------------------------------------------------------------------------------|
| preprocess()     | Any text-cleaning step in other NLP projects                                      |
| build_model()    | Swap the training data or classifier (SVM, Naive Bayes) without changing the flow |
| classify()       | Works with any scikit-learn model that supports predict_proba                     |
| process_ticket() | Called from a web API (Flask/FastAPI), a chatbot or a batch script                |
| TEAM_ROUTES      | Add or rename teams and categories by editing one dictionary                      |

## Possible Improvements

- Move training data to a CSV file and save the trained model with joblib.
- Add unit tests for preprocess and process\_ticket.
- Add logging in place of print statements.